

ENTERPRISE SECURITY BLUEPRINT

ReedShield AI-SPM

Complete Operational Journey & Functional Guide for AI Security Posture Management

Platform Purpose & Non-Technical Overview

ReedShield is an enterprise-grade AI Security Posture Management (AI-SPM) platform designed to discover, govern, and defend AI models, autonomous agents, and RAG context pipelines. This document outlines the step-by-step end-to-end journey through all 13 security operational modules, explaining what each screen displays, what decisions can be made, and how reports can be exported for compliance audits.

APPLICATION VERSION

v3.0.0 (Enterprise Suite)

LIVE ACCESS ENDPOINT

buf-tank-interpretation-elevation.trycloudflare.com

DOCUMENT CLASSIFICATION

CONFIDENTIAL / OPERATIONAL

AS OF DATE

September 2026



Rapid Testing Note: When testing this deployment over the Cloudflare tunnel, all modules feature built-in automated authentication. You land directly inside the live environment without password prompts.

Phase I — Visibility & Executive Cockpit

STEPS 1 TO 3

Step 1: Landing & Automated Zero-Friction Authentication

/admin/dashboard

- ✓ **Access the Link:** Open the provided Cloudflare URL in any desktop or mobile browser.
- ✓ **Auto-Authentication:** The built-in dev-token engine verifies your identity as an enterprise administrator (`spm:admin`) and automatically establishes an active secure session.
- ✓ **Arrival:** You land immediately on the *Executive Posture Dashboard*.

Step 2: Executive Posture Dashboard & Trial Balance

/admin/dashboard

The central command view provides instant executive situational awareness across your entire AI infrastructure.

✓ **Review Trial Balance Metrics:**

KPI Metric	Target / Meaning	Operational Significance
Blended Risk Score	Lower is better (0–100)	Composite posture score calculated across models, agents, and cloud identities.
Open Critical Alerts	Count of unmitigated issues	Immediate attention required (e.g. leaked API keys, exfiltration triggers).
Mean Time to Freeze	Target: < 60 seconds	Automated speed at which an out-of-control agent identity is quarantined.
Modules Enforced	8 of 8 active	Confirmation that guardrails, code scanners, and audit collectors are live.

- ✓ **Inspect High-Confidence Attack Path:** Examine the visual 4-step chain diagram showing how a minor vulnerability (e.g. hardcoded secret) can propagate into a customer PII exfiltration incident.
- ✓ **Export Executive Sheet:** Click the `↓ Export sheet` button located in the top-right of the dashboard card.



Export Action: Generates `reedshield-executive-posture-dashboard-YYYY-MM-DD.csv` with all trial balance metrics and recent critical alerts.

Step 3: AI Inventory & Asset Discovery

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- ✓ **Navigate:** Click **Inventory** in the left sidebar navigation menu.
- ✓ **Switch Tabs:** Toggle between the **Models** tab (tracking Foundation LLMs, Embeddings, Voice models) and the **Agents** tab (tracking LangChain, OpenAI Assistants, AutoGPT instances).
- ✓ **Filter Assets:** Use the dropdown menus to isolate assets by Risk Tier (*Critical, High, Medium, Low*), Cloud Provider (*AWS, Azure, GCP*), or Policy Coverage (*Full, Partial, None*).
- ✓ **Register New Asset:** Click **+ Register Asset** to catalog a newly developed model or agent.
- ✓ **Inspect Asset Details:** Click any asset row to open the lateral drawer showing owners, linked policies, open alerts, and downstream dependencies.



Export Action: Click **↓ Export** to download `reedshield-ai-inventory-models-YYYY-MM-DD.csv` containing model names, providers, owners, and risk status.

Phase II — Security Posture & Threat Detection

STEPS 4 TO 6

Step 4: Posture Gap & Domain Analysis

</admin/posture>

Deep-dive into coverage across all 8 security domains defining operational AI defense:

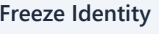
- ✓ **Domain Scores:** Observe real-time evaluations across *Guardrail Enforcement (88/100)*, *Model Integrity (91/100)*, *Data & Knowledge Trust (63/100)*, *Simulation Readiness (54/100)*, and *Identity Scope (79/100)*.
- ✓ **Review Critical Gaps:** Scroll down to the *Posture Gaps* list to read prioritized remediation instructions (e.g. "12 production agents missing strict prompt policy").
- ✓ **Trigger Report Generation:** Click  **Generate Report** to trigger an AI-synthesized gap closure checklist.



Export Action: Click  **Export** to download reedshield-security-posture-YYYY-MM-DD.csv with domain ratings, covered controls, and gap tallies.

Step 5: Threat Findings & Active Security Alerts

</admin/alerts>

- ✓ **Navigate:** Click **Findings** in the left sidebar.
- ✓ **Review Threats:** Review live detection events from the hunt engine, categorized by attack vector (e.g. *Data Exfiltration*, *Prompt Injection*, *Tool Abuse*, *Secret Exposure*).
- ✓ **Investigate a Threat:** Click any finding in the table. The Action Panel opens on the right showing:
 - ✓ The prompt payload and intercepted token stream.
 - ✓ Affected model and agent identities.
 - ✓ One-click mitigation buttons:  **Freeze Identity** ,  **Revoke Token** ,  **Quarantine Tool** .
- ✓ **Add Custom Hunt Rule:** Click  **+ Create Rule** to configure a new behavioral detection threshold.



Export Action: Click  **Export** to download reedshield-threat-findings-YYYY-MM-DD.csv with full incident details and severity classifications.

Step 6: Incident Management & Case Tracking

</admin/cases>

- ✓ **Navigate:** Click **Cases** in the left sidebar.
- ✓ **Coordinate Response:** Track open security investigations, assign team owners, and monitor SLA countdown clocks to prevent regulatory breach notification lapses.
- ✓ **Update Case:** Select a case to add forensic notes, attach evidence payloads, or change status to *Investigating* or *Resolved*.



Export Action: Click **↓ Export** to generate `reedshield-security-cases-YYYY-MM-DD.csv` for incident review meetings.

Phase III — Governance, Runtime & Lineage

STEPS 7 TO 10

Step 7: Data & Knowledge Governance (RAG Safety)

</admin/data>

- ✓ **Inspect Context Sources:** View all vector databases (e.g. Pinecone, Chroma, Milvus), enterprise SharePoint indices, and SQL knowledge sources used by Retrieval-Augmented Generation (RAG) agents.
- ✓ **Check Sensitivity & Trust:** Ensure sensitive repositories (tagged *Restricted* or *Confidential*) maintain a Trust Score ≥ 80 .
- ✓ **Audit PII Exposure:** Identify sources containing unmasked personal identifiers, payment cards, or authentication credentials.

 **Export Action:** Click  to download `reedshield-data-and-knowledge-sources-YYYY-MM-DD.csv`.

Step 8: Policies & Guardrails Configuration

</admin/policies>

- ✓ **Review Rules:** Inspect active guardrails (e.g. *PII-Guard v2*, *Prompt Injection Shield*, *Financial Exfiltration Blocker*).
- ✓ **Toggle Enforcement Mode:** Switch policies between **Monitor** (silent telemetry and logging) and **Enforce** (hard real-time blocking of offending requests).
- ✓ **Audit Waivers:** Manage exceptions and business justifications with automatic expiration dates.
- ✓ **Single Policy Download:** Click the download icon beside any individual policy to download its standalone JSON schema.

 **Export Action:** Click  in the header to download the complete enterprise policy registry in CSV format.

Step 9: Runtime Telemetry & Execution Stream

</admin/runtime>

- ✓ **Monitor Live Traffic:** Watch live AI inferences, external tool invocations, and policy block decisions update in real time.
- ✓ **Stream Controls:** Click  to freeze the screen for inspection; click  to continue streaming.
- ✓ **Filter Suspicious:** Toggle the *Suspicious Only* filter to immediately eliminate benign chatter and isolate flagged actions.

 **Export Action:** Click  to download `reedshield-runtime-execution-events-YYYY-MM-DD.csv`.

Step 10: Context Lineage Graph & Decision Formation

</admin/lineage>

- ✓ **Visual Topology:** View the full graphical execution path showing how an AI decision was produced:

[User Prompt] → [RAG Context Chunk] → [LLM Completion] → [Tool Execution] → [Policy Interceptor]

- ✓ **Replay Session:** Click [▶ Run Simulation](#) to re-execute a historical session's prompt through current guardrails to verify if new policies now block it.



Export Action: Click [↓ Export Trace](#) to download full session graph and node parameters as a JSON report.

Phase IV — Compliance, Shadow AI & Validation

STEPS 11 TO 13

Step 11: Compliance Frameworks Scorecard

/admin/compliance

✓ **Select Framework:** Review control readiness scores across major global standards:

Framework	Category	Target Focus
NIST AI RMF	AI Governance	Govern, Map, Measure, and Manage functions.
ISO/IEC 42001	Management System	Standardized controls for Artificial Intelligence.
EU AI Act	Regulatory / Risk	High-risk AI system compliance and prohibited practices.
DPDP Act, 2023	Privacy / India	Consent, breach notifications, and sensitive personal data.

✓ **Re-Evaluate Controls:** Click  to run real-time evidence queries against your active cloud infrastructure.

 **Export Action:** Click  to download reedshield-compliance-frameworks-YYYY-MM-DD.csv with all control ratings.

Step 12: Shadow AI Discovery & Telemetry

/admin/shadow

- ✓ **Discovered SaaS AI:** Review unmanaged consumer AI applications (e.g. public ChatGPT, Claude, Midjourney, Jasper) detected across corporate networks via DNS and CASB telemetry.
- ✓ **Sanction or Block:** One-click approval to sanction sanctioned corporate subscriptions or block non-compliant platforms.
- ✓ **Data Volume Tracking:** Monitor total MB/GB uploaded to consumer AI to prevent data leaks.

 **Export Action:** Click  to download reedshield-shadow-ai-applications-YYYY-MM-DD.csv.

Step 13: Simulation Lab & Automated Red-Teaming

</admin/simulation>

- ✓ **Pick Attack Vector:** Choose an attack category: *Prompt Injection*, *Jailbreak Attempt*, *Data Exfiltration*, or *Tool Abuse*.
- ✓ **Select Target Agent:** Pick the target agent (e.g. FinanceAssistant-v2) and target model (e.g. gpt-4o).
- ✓ **Compare Mode:** Click ☐ **Compare Mode** to run an attack twice:
 - ✓ **Before:** Demonstrates baseline unprotected agent vulnerability.
 - ✓ **After:** Demonstrates real-time interception by ReedShield policy guardrails.
- ✓ **Run Simulation:** Click to observe step-by-step red-team probes in the live results terminal.